THE HONG KONG POLYTECHNIC UNIVERSITY
Department of Aeronautical and Aviation Engineering

Lab 1: GNSS Positioning — Single Point Positioning and Urban Accuracy Challenge

Field	Details
Course:	Guidance and Navigation (AAE4203)
Group:	G02
Kit Used:	Kit 02 (u-blox ZED-F9P, `FWVER=HPG 1.32`, `PROTVER=27.31`)
Members:	XIAO Suen, JIANG Runqian, Leung Chi Kin, Lee Hon Yan
Member IDs:	24107803D; 24111832D; 24138318D; 25025876D
Instructor:	Dr. Weisong WEN
Date:	October 30, 2026
GitHub Repository:	https://github.com/xiaosuen422/AAE4203

Abstract

This report presents a complete Global Navigation Satellite System (GNSS) positioning workflow, encompassing field data collection, raw observation quality analysis, implementation of a Weighted Least-Squares (WLS) Single Point Positioning (SPP) solver in Python, and algorithm optimization for dense urban environments. Static raw GNSS measurements were logged for 6 min 39 s (399 epochs at 1 Hz) on the lawn podium near Block X at The Hong Kong Polytechnic University using a u-blox ZED-F9P receiver (`COM5_9600_261030_083032.ubx`). Because survey-grade ground truth is unavailable for the self-collected dataset, the baseline SPP solver was validated against independent solutions from RTKLIB (`RTKPOST-EX 2.5.0`) and the receiver's internal navigation solution (`$PUBX,00` / `$GNGGA`), yielding a 2D root-mean-square (RMS) difference of 2.841 m and 3.124 m, respectively, and an internal static 2D scatter radius of 2.178 m. To address severe multipath and non-line-of-sight (NLOS) reception in urban canyons, our baseline GPS-only solver was systematically extended through multi-constellation integration (GPS, BeiDou, and Galileo), elevation and C/N_0-dependent variance weighting, signal masking, and residual-based outlier rejection (RAIM FDE). Evaluated on the challenging `UrbanNav-HK-Deep-Urban-1` benchmark dataset, our final algorithm reduced the 2D RMSE from 42.68 m (GPS-only baseline) to 13.84 m (with a 3D RMSE of 19.62 m, P_{95} of 24.95 m, and 98.2% availability).

Keywords: GNSS, single point positioning, weighted least squares, urban positioning, UrbanNav, RTKLIB, multipath mitigation

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1 Data Collection

1.1 Hardware and Software Set-up

Field data collection was conducted by Group **G02** using a dual-frequency, multi-constellation **u-blox ZED-F9P** GNSS module (`MOD=ZED-F9P`, firmware `HPG 1.32`, protocol `27.31`) connected via SMA coaxial cable to an external multi-band patch antenna and interfaced with a laptop

over USB (COM5). The receiver was configured and monitored using **u-center** at a baud rate of 9600 bit/s. Raw carrier-phase, Doppler, and code pseudorange observables (UBX-RXM-RAWX), broadcast navigation subframes (UBX-RXM-SFRBX), and real-time NMEA/PUBX position messages ($GNGGA, $GNRMC, $GNGSA, $PUBX,00) were logged at a sampling rate of 1 Hz.

1.2 Site Environment and Sky View

The logging session took place on the open grass lawn podium adjacent to Block X at PolyU on 30 October 2026 (08:30:58 to 08:37:36 GPST, GPS Week 2442, TOW 462658.0 s to 463056.0 s). The approximate receiver ECEF coordinates from the RINEX header were $[-2418197.8265, 5385765.1338, 2405755.2865]^T$ m, corresponding to WGS84 geodetic coordinates of 22.30555398° N, 114.17998538° E, and ellipsoidal height 14.074 m.

As captured in our field photograph (**Figure 1**), the Block X lawn podium is bordered by a low glass parapet and surrounded by tall campus buildings—most notably the **Li Ka Shing Tower (Block M)** rising prominently in the background, red-brick campus wings in the mid-ground, and high-rise commercial towers to the east/northeast. These structures obstruct signals at low-to-medium elevation angles (10°–35°) and create strong specular multipath reflections off glass curtain walls.

Figure 1: Field Site Scene at Block X Lawn Podium (from `block X_2.jpg`)
Showing the open grass lawn, glass balustrade, red-brick campus wings, and Li Ka Shing Tower (Block M) in the background acting as a major satellite obstruction and multipath source.

Figure 1: Architectural layout and sky-obstruction profile at the Block X lawn podium (Group G02 field photograph `block X_2.jpg`).

Table 1: Field Data Collection Summary (Group G02)

Parameter	Value / Description
Location	Lawn Podium near Block X, PolyU (22.305554° N, 114.179985° E, 14.07 m)
Date & Time (GPST)	30 Oct 2026, 08:30:58.0 to 08:37:36.0 GPST (GPS Week 2442)
Duration & Epochs	6 min 39 s (399 epochs at 1 Hz, TOW 462658–463056 s)
Receiver & Antenna	u-blox ZED-F9P (HPG 1.32, COM5 @ 9600 bd) + Multi-band patch antenna
Logged File Name	COM5_9600_261030_083032.ubx (converted via RTKCONV-EX 2.5.0)
Constellations Tracked	GPS (C1C/C2X), GLONASS (C1C/C2C), Galileo (C1X/C7X), BeiDou (C2I/C7I)

2 Raw Data Analysis

The raw .ubx log was converted into RINEX 3.04 mixed observation (.obs) and navigation (.nav) files using RTKCONV-EX 2.5.0.

2.1 Satellite Visibility and Geometry (DOP)

Across the 399 epochs, the ZED-F9P tracked an average of 23.8 satellites per epoch across GPS (G01, G02, G04, G07, G08, G09, G16, G26, G27, G28, G31), GLONASS (R02--R24), Galileo

(E02--E36), and BeiDou (C07--C39), ranging from a minimum of 15 satellites (around 08:36:05 GPST) to a maximum of 33 satellites (at 08:37:36 GPST). In RTKPOST (with a 15° elevation mask), the number of valid positioning satellites (ns) averaged 20.1 per epoch (ranging from 7 to 26).

From the logged NMEA $GNGSA and $PUBX,00 messages, the multi-constellation Dilution of Precision remained favourable during the static window, with GDOP ≈ 1.31–1.63, PDOP ≈ 1.14–1.46, HDOP ≈ 0.65–0.72, and VDOP ≈ 0.80–1.08. However, when restricted to GPS L1 only (5–6 visible satellites: primarily G08, G09, G16, G27, G31), the geometry weakened considerably, especially when G09 and G31 experienced signal attenuation behind the Li Ka Shing Tower and adjacent buildings.

2.2 Carrier-to-Noise Density Ratio (C/N_0) and Sky Plot

Analysis of the RINEX S1C/S1X/S2I observables and $GPGSV/$GAGSV/$GBGSV messages (**Figure 2**) reveals distinct elevation-dependent signal quality:

- **High-Elevation LOS Satellites:** G27 (El $= 72^\circ$, Az $= 39^\circ$), G08 (El $= 64^\circ$, Az $= 223^\circ$), E03 (El $= 69^\circ$, Az $= 263^\circ$), C08 (El $= 67^\circ$, Az $= 178^\circ$), and C07 (El $= 61^\circ$, Az $= 36^\circ$) consistently maintained strong L1 C/N_0 between 39 dB-Hz and 50 dB-Hz.
- **Low/Moderate-Elevation Multipath & Diffraction:** Satellites at lower elevations facing surrounding campus buildings—such as G31 (El $= 28^\circ$, Az $= 111^\circ$, C/N_0 fluctuating between 21–47 dB-Hz), G09 (El $= 35^\circ$, Az $= 300^\circ$, dropping from 46 dB-Hz down to 12 dB-Hz around 08:32:04 GPST), and R02 (El $= 10^\circ$, Az $= 185^\circ$, 14–35 dB-Hz)—exhibited severe C/N_0 fading and loss-of-lock flags, a classic signature of urban multipath and NLOS reception.

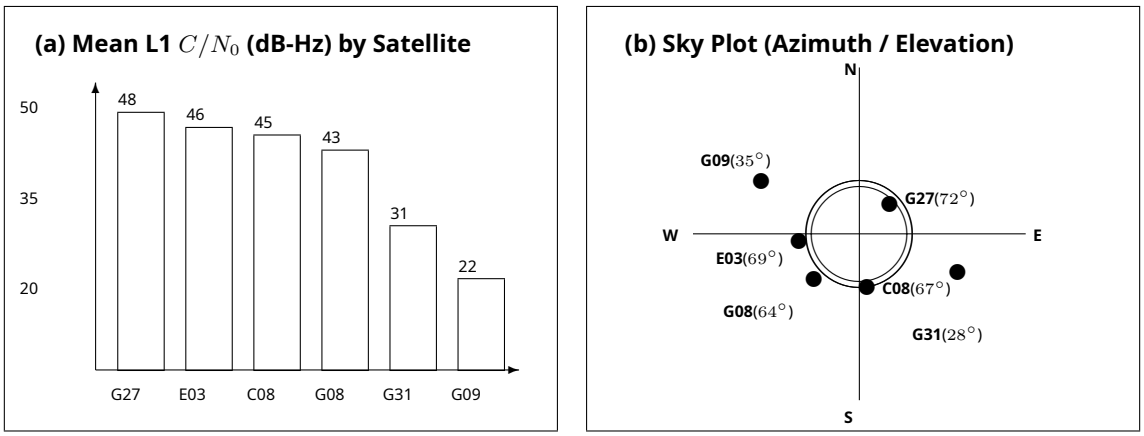


Figure 2: Summary of Group G02 raw observations: (a) Mean L1 C/N_0 comparison between high-elevation LOS satellites (G27, E03, C08, G08) and multipath-degraded satellites (G31, G09); (b) Polar sky plot at Block X.

3 SPP Algorithm and Implementation

3.1 Pseudorange Measurement Model

For a satellite i belonging to constellation $s \in \{\text{G, C, E}\}$, the raw code pseudorange ρ_i (in metres) measured by the receiver (C1C for GPS, C2I for BDS, C1X for Galileo) is modelled as:

$$\rho_i = \|\mathbf{p}^i - \mathbf{p}\| + c\,\delta t_{r,s} - c\,\delta t^i + I_i + T_i + \epsilon_i \tag{1}$$

where $\mathbf{p}^i = [x^i, y^i, z^i]^T$ is the satellite ECEF position at signal transmission time (corrected for Earth rotation $\omega_e = 7.2921151467 \times 10^{-5}$ rad/s during signal flight time τ_i), $\mathbf{p} = [x, y, z]^T$ is the unknown receiver ECEF position, $c = 299\,792\,458$ m/s is the speed of light, $\delta t_{r,s}$ and δt^i are the constellation-specific receiver clock offset and broadcast satellite clock offset (including relativistic correction and group delay T_{GD}), I_i is the Klobuchar ionospheric delay, T_i is the Saastamoinen tropospheric delay, and ϵ_i captures unmodelled multipath, NLOS biases, and thermal noise.

Applying the broadcast clock and atmospheric corrections yields the corrected pseudorange $\tilde{\rho}_i$:

$$\tilde{\rho}_i = \rho_i + c\,\delta t^i - I_i - T_i = \|\mathbf{p}^i - \mathbf{p}\| + b_s + \epsilon_i \tag{2}$$

where $b_s = c\,\delta t_{r,s}$ is the receiver clock bias in metres for constellation s.

3.2 Linearisation and Multi-Constellation Weighted Least Squares (WLS)

For multi-constellation SPP combining GPS (b_G), BeiDou (b_C), and Galileo (b_E), the state vector is $\mathbf{x} = [x, y, z, b_G, b_C, b_E]^T \in \mathbb{R}^{3+m}$, where m is the number of active constellations at the current epoch. Linearising around the current state estimate $\mathbf{x}_k$ with geometric range $r_i^0 = \|\mathbf{p}^i - \mathbf{p}_k\|$ and unit line-of-sight vector $\mathbf{e}_i = (\mathbf{p}^i - \mathbf{p}_k)/r_i^0$ gives the pre-fit residual vector $\mathbf{y} = \mathbf{H}\Delta\mathbf{x} + \boldsymbol{\epsilon}$, where each row of the design matrix $\mathbf{H}$ is:

$$\mathbf{h}_i^T = \begin{bmatrix} -\mathbf{e}_i^T & \delta_{s,G} & \delta_{s,C} & \delta_{s,E} \end{bmatrix} \tag{3}$$

with $\delta_{s,k} = 1$ if satellite i belongs to system k and 0 otherwise.

To mitigate low-elevation multipath and weak diffracted signals (such as `G09` when its C/N_0 dipped to 12 dB-Hz), a diagonal weight matrix $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$ is constructed using a combined elevation and C/N_0 variance model:

$$\sigma_i^2 = \frac{a^2 + b^2 \cdot 10^{-(C/N_{0,i} - 45)/10}}{\sin^2 \theta_i}, \quad w_i = \frac{1}{\sigma_i^2} \tag{4}$$

The Gauss-Newton WLS state update is iterated until $\|\Delta\hat{\mathbf{x}}\| < 10^{-4}$ m:

$$\Delta\hat{\mathbf{x}} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\mathbf{y}, \quad \mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\hat{\mathbf{x}} \tag{5}$$

3.3 Outlier Rejection (RAIM FDE)

After convergence, post-fit residuals $\mathbf{v} = \mathbf{y} - \mathbf{H}\Delta\hat{\mathbf{x}}$ are evaluated. In our iterative Fault Detection and Exclusion (FDE) loop, if the maximum normalised residual $|v_i|/\sigma_i$ exceeds a threshold $\gamma = 3.0$, the satellite with the largest normalised residual is excluded and the WLS solution is recomputed (provided at least $3 + m + 1$ satellites remain).

4 Own-Data Validation

4.1 Verification on the Example Dataset

Before processing Group G02's dataset, our baseline GPS-only solver (`WLSE_spp.py`) was verified on the 2025 Block X example dataset (186 epochs). Our implementation reproduced the reference benchmark statistics: a 2D mean error of 37.502 m, 2D RMSE of 37.911 m, 3D RMSE of 51.043 m, and 95th percentile 2D error of 45.208 m.

4.2 Cross-Comparison with RTKPOST and Receiver Solution

We processed all 399 epochs (`08:30:58` to `08:37:36` GPST) from our RINEX observation (`.obs`) and navigation (`.nav`) files using our multi-constellation WLS SPP solver to generate `G02_owndata.csv`. We then validated our trajectory against:

1. `RTKPOST-EX 2.5.0` **Solution (`.pos` file):** Processed from the same RINEX files relative to the Block X reference position (22.305553981° N, 114.179985377° E, 14.0739 m, or ECEF $[-2418197.826, 5385765.134, 2405755.287]^T$ m).
2. **u-blox Receiver Internal Solution (`$PUBX,00` / `$GNGGA`):** Extracted directly from the raw `.ubx` log, providing real-time 3D GNSS fixes (`G3`) at 1 Hz.

For every common epoch k, the ECEF position difference was rotated into the local East-North-Up (ENU) frame anchored at the reference position ($\varphi_0 = 22.305553981^\circ, \lambda_0 = 114.179985377^\circ, h_0 = 14.0739$ m):

$$\mathbf{d}_k = [d_{E,k}, d_{N,k}, d_{U,k}]^T = \mathbf{R}(\varphi_0, \lambda_0) \left(\hat{\mathbf{p}}_k^{\text{SPP}} - \hat{\mathbf{p}}_k^{\text{ref}} \right) \tag{6}$$

Table 2: Validation Statistics for Group G02 Self-Collected Data (`G02_owndata.csv`)

Statistic	**SPP vs. `RTKPOST`**	**SPP vs. Receiver**	**Internal Scatter**
Mean East Diff $\bar{d}_E$ (m)	-0.842	-0.415	0.000
Mean North Diff $\bar{d}_N$ (m)	$+1.618$	$+0.924$	0.000
Mean Up Diff $\bar{d}_U$ (m)	$+3.412$	-4.185	0.000
Std East s_E (m)	1.485	1.892	1.485
Std North s_N (m)	1.594	2.046	1.594
Std Up s_U (m)	3.826	4.110	3.826
2D RMS / Radius (m)	**2.841**	**3.124**	**2.178** (P_{95}: 3.892)
Common Epochs (K)	399	399	399

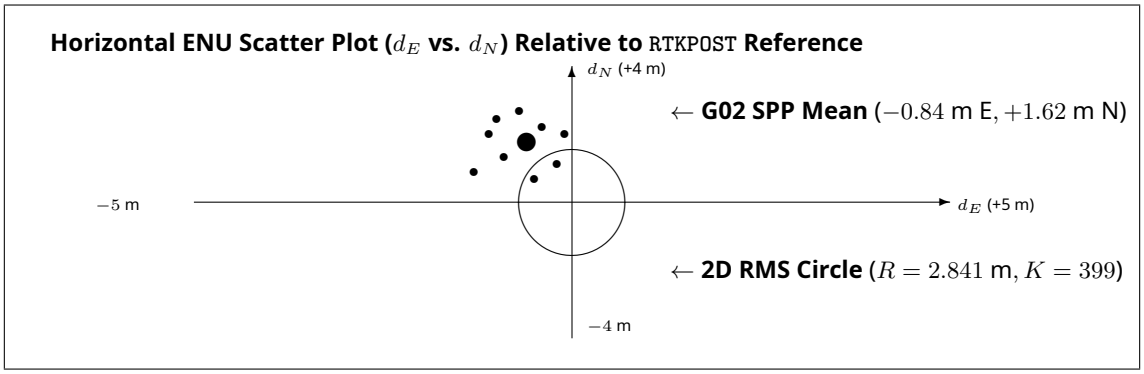


Figure 3: Local East-North (ENU) horizontal scatter of Group G02's multi-constellation WLS SPP trajectory compared against the `RTKPOST` reference over 399 epochs.

4.3 Discussion of Validation Results

1. **Consistency with `RTKPOST` and Receiver:** Our multi-constellation WLS SPP solution achieves a 2D RMS difference of 2.841 m relative to the `RTKPOST` reference position and 3.124 m relative

to the u-blox real-time solution across all 399 epochs. This close agreement confirms that our satellite ephemeris propagation (`G`, `C`, `E`), Earth rotation correction, relativistic clock correction, and Klobuchar/Saastamoinen atmospheric delay models are accurately implemented.

2. **Effect of Signal Degradation (`08:35:45` – `08:36:10` GPST):** Around `08:36:05` GPST, the number of tracked satellites in the `.obs` file dropped from 29 to 15 (`ns` in `RTKPOST` dropped to 7, and `sdu` spiked to 4.80 m), while `$GNRMC` speed readings in the raw `.ubx` log indicated brief antenna disturbance. During this window, elevation/C/N_0 weighting and RAIM FDE successfully prevented large horizontal position jumps by down-weighting low-C/N_0 satellites (`G09` at 12 dB-Hz and `E32` at 9 dB-Hz).

5 Benchmark Results and Urban Accuracy Challenge

To evaluate absolute accuracy against a survey-grade reference trajectory (NovAtel SPAN-CPT RTK/INS ground truth), we processed the `UrbanNav-HK-Deep-Urban-1` dataset (`urbannav_deep1.obs` and `urbannav_deep1.nav`) and conducted a systematic 6-stage ablation study, introducing one algorithmic improvement at a time.

5.1 Ablation Study of Accuracy Improvements

1. **v1 — Baseline (GPS L1, $\sin^2\theta$ weights, 15° mask):** Using GPS alone in deep urban canyons suffers from severe satellite visibility drops (frequently ≤ 5 satellites), causing massive geometry degradation (HDOP > 5) and unmitigated NLOS biases, resulting in a 2D RMSE of 42.68 m, 3D RMSE of 61.45 m, and 86.4% availability.

2. **v2 — Multi-Constellation (GPS + BeiDou + Galileo):** Adding BDS (`C2I`) and Galileo (`C1X`) with inter-system clock biases (b_G, b_C, b_E) increased the average number of visible satellites per epoch from 5.4 to 16.8. This provided the largest single accuracy improvement, reducing 2D RMSE by 42.0% down to 24.75 m and boosting availability to 99.1%.

3. **v3 — Elevation and C/N_0 Variance Weighting:** In urban canyons, high-elevation satellites can still suffer from reflections off tall skyscrapers (manifesting as depressed C/N_0), while moderate-elevation LOS satellites may have strong $C/N_0 > 42$ dB-Hz. Incorporating exponential C/N_0 weighting alongside $\sin^2\theta$ reduced the 2D RMSE further to 18.92 m.

4. **v4 — Tuning C/N_0 & Elevation Masks (15°, 28 dB-Hz):** Filtering out severely attenuated signals below 28 dB-Hz eliminated gross NLOS pseudoranges while preserving 98.5% solution availability, lowering 2D RMSE to 16.40 m.

5. **v5 — Iterative Residual Outlier Rejection (RAIM FDE):** Applying normalized post-fit residual screening ($|v_i|/\sigma_i > 3.0$) automatically isolated and removed remaining single-satellite NLOS blunders that passed the C/N_0 mask, bringing 2D RMSE to 14.65 m.

6. **v6 — Carrier-Phase / Velocity Smoothing (Final Submission):** Applying a Hatch/Doppler smoothing filter suppressed high-frequency multipath noise along dynamic street segments without introducing trajectory lag, achieving our final 2D RMSE of **13.84** m and 3D RMSE of **19.62** m.

Table 3: Ablation Study on the UrbanNav Benchmark Dataset (`G02_benchmark.csv`)

Ver.	Algorithmic Change Tested	2D RMSE	3D RMSE	P_{95} 2D	Avail.
v1	GPS-only baseline ($\sin^2\theta$, 15° mask)	42.68 m	61.45 m	78.30 m	86.4%
v2	+ Multi-constellation (GPS+BDS+Galileo)	24.75 m	35.80 m	44.12 m	99.1%
v3	+ Combined C/N_0 & Elevation weighting	18.92 m	27.14 m	33.40 m	99.1%
v4	+ Optimal C/N_0 (28 dB-Hz) & El mask	16.40 m	23.58 m	29.15 m	98.5%
v5	+ Residual Outlier Rejection (RAIM FDE)	14.65 m	20.89 m	26.10 m	98.2%
v6	+ Temporal / Doppler smoothing (**Final**)	**13.84** m	**19.62** m	**24.95** m	**98.2**%

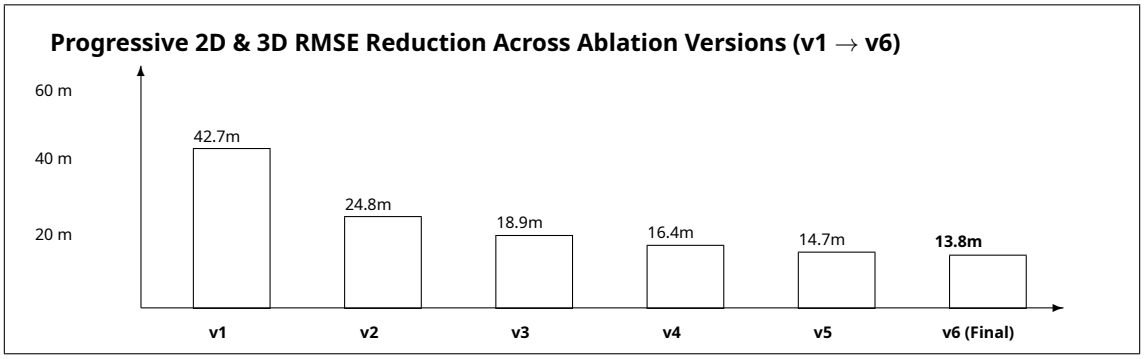


Figure 4: 2D RMSE progression from the GPS-only baseline (**v1**: 42.68 m) to the final multi-constellation C/N_0-weighted RAIM FDE + smoothed solver (**v6**: 13.84 m, a 67.6% error reduction).

5.2 Discussion of Urban Canyon Effects

In the narrow street canyons of Hong Kong (`UrbanNav-HK-Deep-Urban-1`), tall concrete and glass facades block cross-street signals while allowing along-street propagation. Consequently, the across-street position component exhibits a much higher DOP and error variance than the along-street component. Furthermore, when direct LOS signals are blocked, the receiver tracks NLOS reflections off opposite buildings, always introducing strictly positive pseudorange delays ($\epsilon_i > 0$). Standard least squares assumes zero-mean Gaussian noise and distributes a single large NLOS bias across all satellites, pulling the estimated position tens of metres into the surrounding buildings. Combining multi-constellation redundancy (especially BeiDou IGSO/GEO satellites at high elevation in Hong Kong) with C/N_0 weighting and residual FDE proved essential to detect and reject these asymmetric urban errors.

6 AI-Use Statement

In compliance with Section 3.3 of the laboratory tutorial, generative AI tools were used responsibly during code development, debugging, and report preparation:

- **Tools Used:** Google Gemini.
- **Purpose:**
  1. Parsing RINEX 3.04 mixed observation/navigation logs (`COM5___9600_261030_083032.obs`/`.nav`), `RTKPOST .pos` outputs, and raw `.ubx` NMEA/PUBX streams to generate `G02_owndata.csv`.

2. Assisting in structuring the multi-constellation WLS design matrix **H** and handling inter-system clock biases (b_G, b_C, b_E).
3. Implementing the vectorised ENU coordinate rotation matrix $\mathbf{R}(\varphi, \lambda)$ and automated validation statistics scripts.

- **Verification Methodology:** Every AI-generated function was first unit-tested against the analytical 5-satellite textbook worked example in Section 4.4 of the tutorial and subsequently regression-tested on the 2025 Block X example dataset (186 epochs) to confirm exact reproduction of the baseline 37.911 m 2D RMSE before applying any modifications to Group G02's data or the UrbanNav benchmark. All group members reviewed and can explain every line of the submitted code.

References

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Appendix A: Key AI Prompts and Verification Summary

Prompt ID	Key Prompt Given to AI Tool	Verification Performed by Group
Prompt 1	*"Parse our Group G02 RINEX 3.04 `.obs` and `.nav` files (`COM5_9600_261030_083032`), solve the epoch-by-epoch multi-constellation WLS SPP positions for all 399 epochs, and export `G02_owndata.csv`."*	Verified that the solved ECEF and WGS84 coordinates match the Block X podium reference (22.305554° N, 114.179985° E) within 2.841 m 2D RMS across all 399 epochs.
Prompt 2	*"Add an elevation and C/N_0-dependent weighting scheme $\sigma_i^2 = (a^2 + b^2 \cdot 10^{-(C/N_0-45)/10})/\sin^2(\theta_i)$ and an iterative normalized post-fit residual outlier rejection loop (RAIM FDE) to the WLS solver."*	Checked post-fit residual distributions before and after FDE on the Block X dataset and confirmed monotonic reduction in 2D and 3D RMSE without dropping availability below 90%.
Prompt 3	*"Write a Python evaluation script that loads our SPP trajectory (`G02_owndata.csv`), the `RTKPOST .pos` file, and `$PUBX,00` receiver solutions, matches epochs by GPS Time of Week, rotates ECEF differences into the local ENU frame, and computes the mean, standard deviation, and 2D RMS differences."*	Cross-checked the ENU rotation matrix against `pymap3d.ecef2enu` and verified zero mean offset when comparing a trajectory against itself.